

Short-term blackcurrant extract consumption modulates exercise-induced oxidative stress and lipopolysaccharide-stimulated inflammatory responses.



Exercise-induced oxidative stress is instrumental in achieving the health benefits from regular exercise. Therefore, inappropriate use of fruit-derived products (commonly applied as prophylactic antioxidants) may counteract the positive effects of exercise. Using human exercise and cellular models we found that 1) blackcurrant supplementation suppressed exercise-induced oxidative stress, e.g., plasma carbonyls (0.9 $\pm$ 0.1 vs. 0.6 $\pm$ 0.1 nmol/mg protein, placebo vs. blackcurrant), and 2) preincubation of THP-1 cells with an anthocyanin-rich blackcurrant extract inhibited LPS-stimulated cytokine secretion [TNF- α (16,453 $\pm$ 322 vs. 10,941 $\pm$ 82 pg/ml, control vs. extract, $P < 0.05$) and IL-6 (476 $\pm$ 14 vs. 326 $\pm$ 32 pg/ml, control vs. extract, $P < 0.05$)] and NF-kappaB activation. In addition to its antioxidant and anti-inflammatory properties, we found that postexercise plasma collected after blackcurrant supplementation enhanced the differential temporal LPS-stimulated inflammatory response in THP-1 cells, resulting in an early suppression of TNF- α (1,741 $\pm$ 32 vs. 1,312 $\pm$ 42 pg/ml, placebo vs. blackcurrant, $P < 0.05$) and IL-6 (44 $\pm$ 5 vs. 36 $\pm$ 3 pg/ml, placebo vs. blackcurrant, $P < 0.05$) secretion after 24 h. Furthermore, by using an oxidative stress cell model, we found that preincubation of THP-1 cells with hydrogen peroxide (H₂O₂) prior to extract exposure caused a greater suppression of LPS-stimulated cytokine secretion after 24 h, which was not evident when cells were simultaneously incubated with H₂O₂ and the extract. In summary, our findings support the concept that consumption of blackcurrant anthocyanins alleviate oxidative stress, and may, if given at the appropriate amount and time, complement the ability of exercise to enhance immune responsiveness to potential pathogens.